

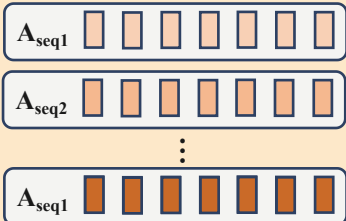
Admissibility Rule Extraction

Expert Trajectory

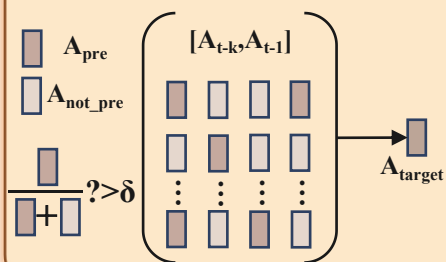
"taskName": "mendelian-genetics-unknown-plant"
 "taskDescription": "Your task is to determine whether blue ..."
 "path": [{"action": "look around", "observation": "This room is called the hallway. In it, you see ..."}]

Neural-Symbolic Trace Parsing

Action Seq Set

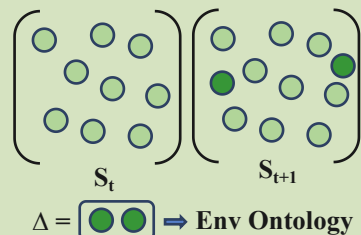


Behavioral Dependency Miner

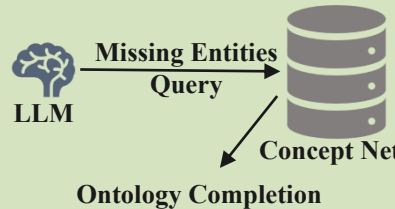


Behavioral Ontology (O_{beh})

Environmental Causality Inducer

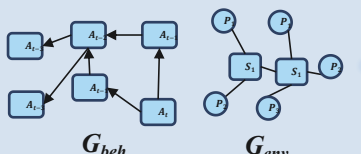


Hybrid Knowledge Completion



Environmental Ontology (O_{env})

Constraint Graph Construction



Constraint-Aware Planning



LLM Agent Planner

Observation (O_t)
 Task description
 Historical behavior
 Additional Info

Generated Candidate Action $\hat{a}_t$

Constraint-Verification

$v_{beh} \leftarrow \text{CheckHistory}(\hat{a}_t, \mathcal{H}, G_{beh})$
 $v_{env} \leftarrow \text{CheckState}(\hat{a}_t, S_t, G_{env})$
 $\text{Valid} \leftarrow v_{beh} \text{ and } v_{env}$

Adaptive Prompt Synthesizer

Observation (O_t)

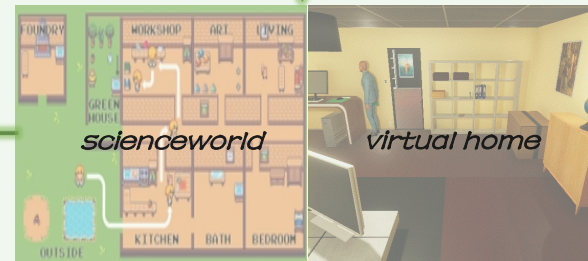
Action Refinement & Re-prompting

Iteration Verification

Iteration not overflow

Iteration limit Exceeded

Fail



Environment